

MODULE 08

Pelvis & Hip CT: Fracture Classification Systems

Young-Burgess · Tile · Garden · Pauwels · Denis · Pipkin · Judet-Letournel · Schatzker

1.0 AMA PRA Cat 1

Pelvis

CME Article Review

LEARNING OBJECTIVES

1. Apply Young-Burgess classification to pelvic ring fractures (LC, APC, VS, CM).
2. Grade femoral neck fractures using the Garden (displacement) and Pauwels (orientation) systems.
3. Apply Denis classification to sacral fractures (Zones I–III).
4. Classify femoral head fractures post-dislocation using the Pipkin system.
5. Identify complex acetabular fracture patterns using the Judet-Letournel classification.
6. Apply Schatzker tibial plateau fracture classification on CT.

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1. Young-Burgess Pelvic Ring Fracture Classification

Type	Subtype	Key Features	Stability / Hemorrhage
LC	I	Sacral buckle Fx; ipsilateral pubic rami Fx (horizontal)	Stable; low hemorrhage
LC	II	LC-I + ipsilateral iliac crescent (posterior iliac wing) Fx	Partially unstable
LC	III	LC-II + contralateral open-book pattern	Highly unstable; MAJOR hemorrhage
APC	I	Symphyseal diastasis <2.5 cm; ligaments intact	Stable
APC	II	Diastasis >2.5 cm; anterior SI torn; posterior SI intact	Partially unstable; moderate hemorrhage
APC	III	Complete SI disruption; all SI ligaments torn	MOST unstable; major hemorrhage
VS	—	Vertical displacement; SI or sacral disruption; Malgaigne Fx	Highly unstable; major hemorrhage

CRITICAL

APC-III and VS carry the highest life-threatening hemorrhage risk from internal iliac vessels. Emergent pelvic binder and angioembolization for hemodynamic instability.

2. Femoral Neck — Garden & Pauwels Classifications

Garden Classification — Displacement

Stage	Description	AVN Risk	Management
I	Incomplete/impacted; valgus; no displacement	~5%	Fixation or conservative
II	Complete; NO displacement; trabeculae aligned	~10%	Cannulated screw fixation
III	Complete; PARTIAL displacement; trabecular misalignment	~25–35%	Fixation (young); THA (elderly)
IV	Complete; FULL displacement; head freely rotated	~35–50%	THA (elderly); hemi/fixation (young)

Pauwels Classification — Fracture Orientation & AVN Risk

Type	Angle from Horizontal	Forces	AVN Risk
I	<30° (near horizontal)	Compressive — most stable	Lowest

Type	Angle from Horizontal	Forces	AVN Risk
II	30–50° (intermediate)	Mixed compressive/shear	Moderate
III	>50° (near vertical)	SHEAR — most unstable; highest non-union/AVN	HIGHEST

3. Denis Sacral Zones, Pipkin & Schatzker

Denis Sacral Fracture Zones

Zone	Location	Nerve Risk	Implication
I — Alar	Lateral to sacral foramina	~6% (L5)	Usually stable; ORIF if ring disruption
II — Transforaminal	Through the sacral foramina	~28% (S1–S3)	Bladder/bowel risk; ORIF; watch for root injury
III — Central	Through the spinal canal	~57% (cauda equina)	Highest neurological risk; neurosurgical involvement

Pipkin Femoral Head Fractures (Post-Dislocation)

Type	Fragment	Prognosis	Management
I	Inferior to fovea (non-WB)	Better	Closed reduction; fragment excision if needed
II	Superior to fovea (WB dome)	Worse	ORIF if >1 cm or displaced
III	Head + femoral NECK fracture	High AVN rate	Urgent fixation; THA in elderly
IV	Head + ACETABULAR fracture	Complex	ORIF acetabulum; head per size

Schatzker Tibial Plateau Fractures

Type	Pattern	Key Feature	ORIF?
I	Lateral plateau pure SPLIT	Vertical cleavage; no depression	Yes if displaced >3–5 mm
II	Lateral SPLIT + DEPRESSION	Split + central impaction	Yes — elevate and graft
III	Lateral pure DEPRESSION	Central impaction; lateral cortex intact	Yes — elevate sub-articular fragment
IV	MEDIAL plateau	High-energy; peroneal nerve risk	Yes — varus unstable

Type	Pattern	Key Feature	ORIF?
V	BICONDYLAR; no shaft dissociation	Both plateaus involved	Yes — complex
VI	BICONDYLAR + METAPHYSEAL DISSOCIATION	Complete shaft-joint separation	Yes — highest complexity

References

- Young JW, Burgess AR. Radiologic Management of Pelvic Ring Disruptions. 1987.
- Garden RS. J Bone Joint Surg Br 1961.
- Denis F. Spine 1983.
- Schatzker J et al. Clin Orthop 1979.